Sustainable Electronics and E-Waste Management

- Problem/Solution Presentation Draft
- Armani Tapp

Introduction

- E-waste is one of the fastest-growing waste streams in the world.
- Improper disposal releases harmful materials and wastes valuable resources.
- This presentation examines causes, impacts, survey findings, and solutions.

The Problem

- Rapid device replacement increases electronic waste.
- Many consumers lack recycling awareness.
- Millions of devices remain stored or improperly discarded.

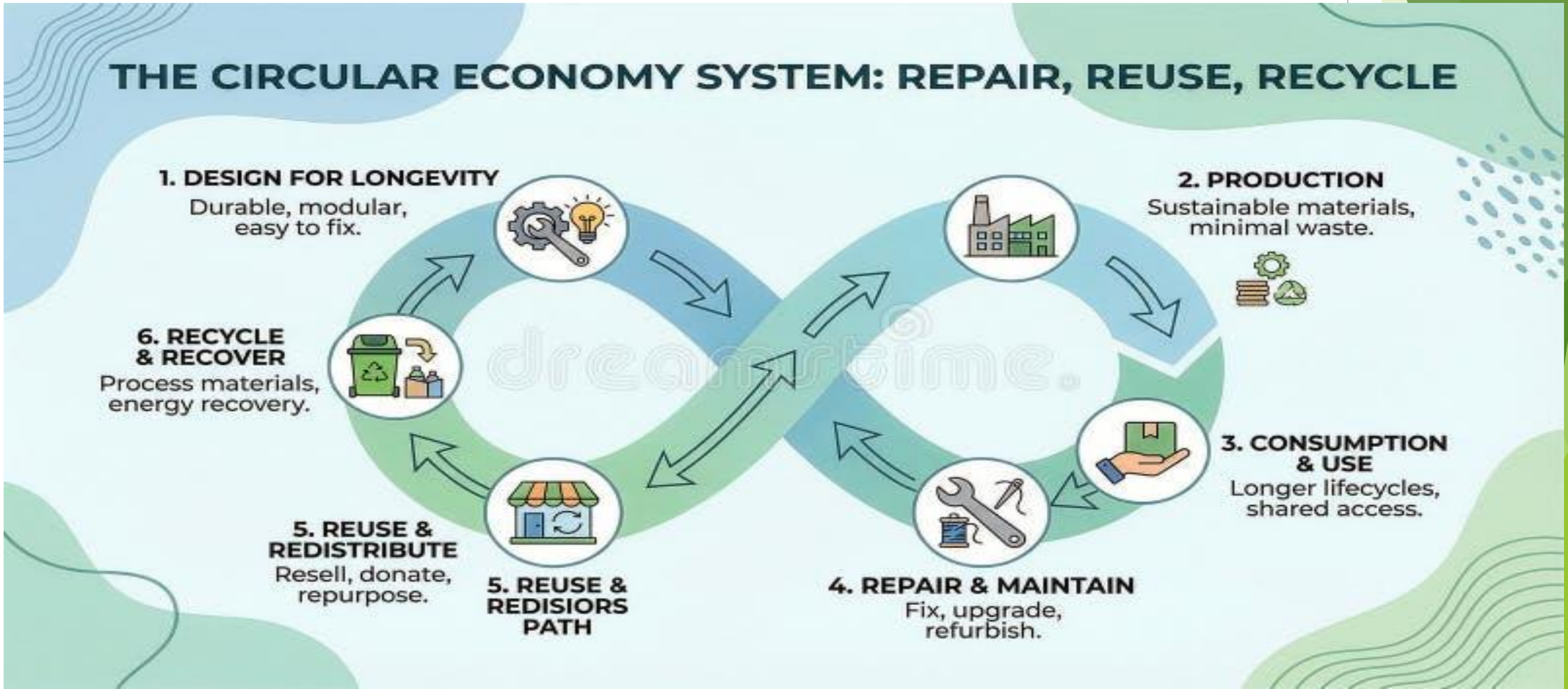
Environmental and Economic Impact

- Toxic materials can contaminate soil and water.
- Recycling recovers valuable metals and reduces resource extraction.
- Sustainable management supports a circular economy.

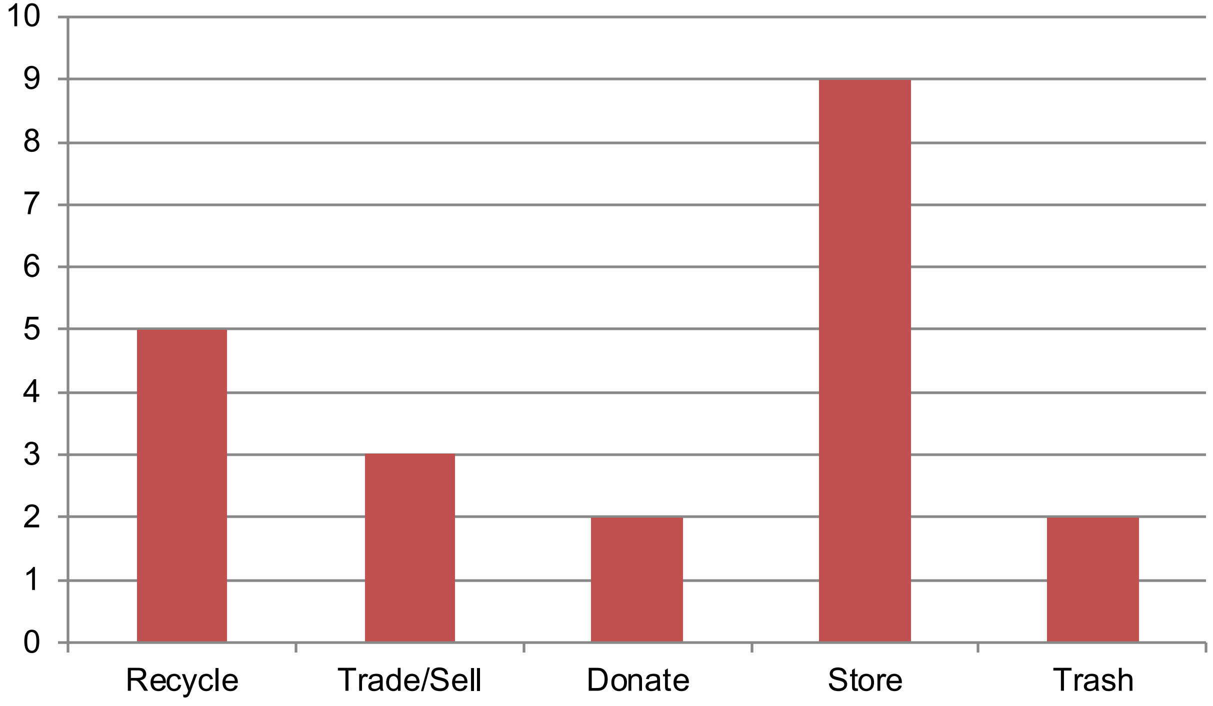
Recycling electronics helps recover valuable materials and reduces environmental harm (Peng et al., 2025). Sustainable management of electronics reduces waste and conserves natural resources (U.S. Environmental Protection Agency[EPA], 2026).

Survey Methodology

- 21 participants surveyed.
- Audience included classmates, friends, family, and coworkers.
- Survey distributed through social media, email, and direct messages.

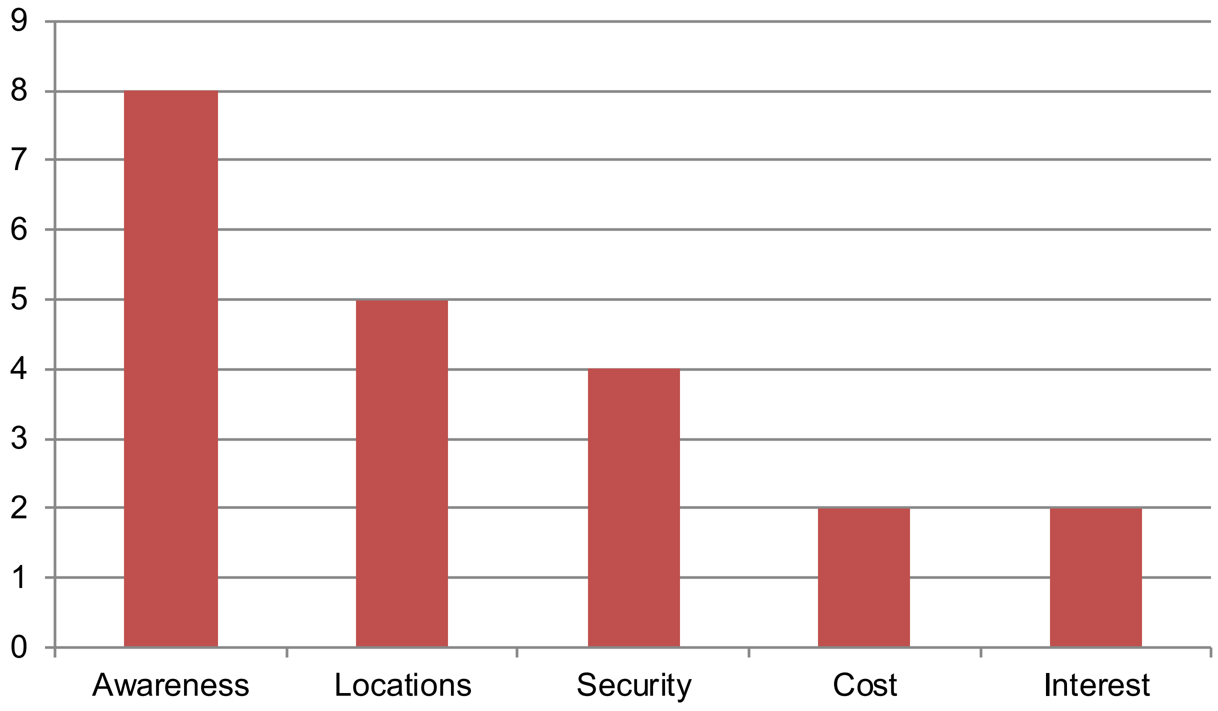


Survey Results: Disposal Habits



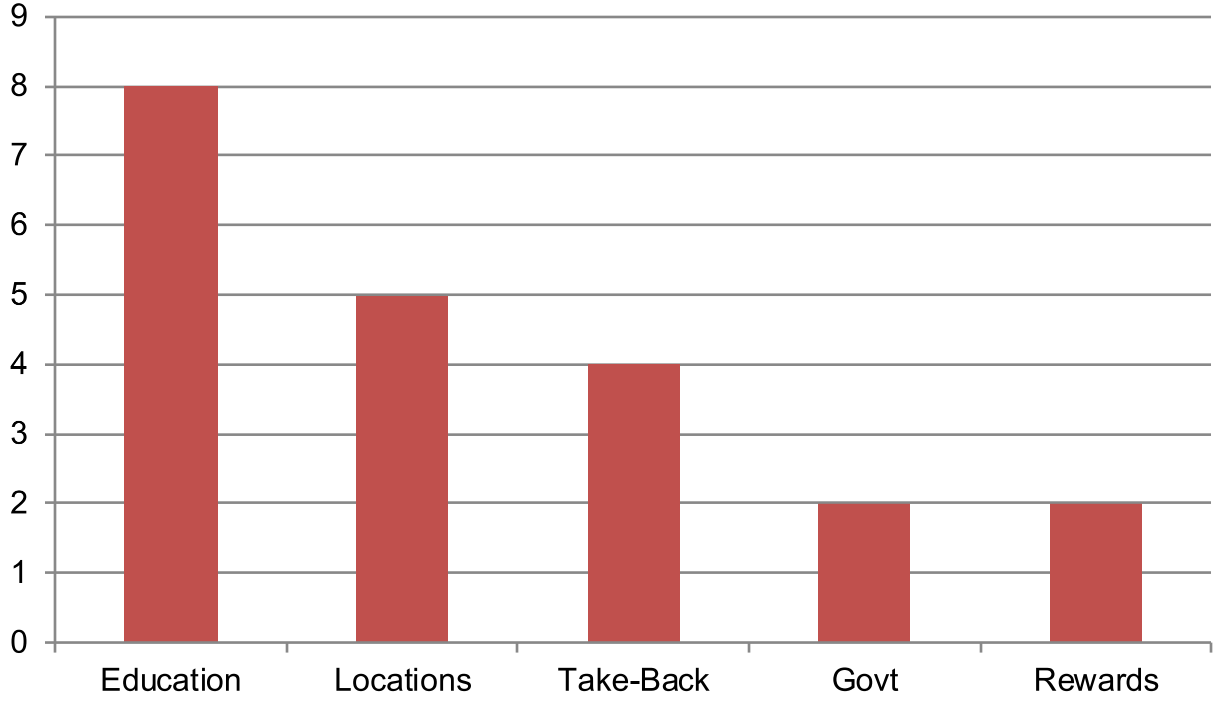
Analysis:
43% of respondents store old electronics at home. This suggests many devices never reach recycling programs.

Survey Results: Recycling Barriers



Analysis:
Lack of awareness was the most common barrier, supporting education campaigns.

Survey Results: Preferred Solutions



Analysis:
Education campaigns received the most support, followed by convenient recycling options.

Proposed Solutions

- Public education and awareness campaigns.
- Convenient electronics recycling locations.
- Manufacturer take-back programs.
- Government incentives and regulations.

Conclusion

- Survey results show awareness is a major challenge.
- Many people store unused electronics at home.
- Education, accessibility, and producer responsibility can reduce e-waste.

References

Quinto, S., et al. (2025). Exploring the E-Waste Crisis. Recycling.

Peng, Y., et al. (2025). E-Waste Take-Back and Producer Eco-innovation. Production and Operations Management.

U.S. Environmental Protection Agency. (2026). Electronics Basic Information, Research, and Initiatives.